

RESEARCH, DESIGN, AND USER EXPERIENCE EVALUATION OF THE VNEID SYSTEM USING FIGMA AND FRAMER TOWARDS USER EXPERIENCE ENHANCEMENT

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1. Introduction

Within the framework of Vietnam's National Digital Transformation Program toward 2025, the VNeID application - developed by the National Population Data Center under the Ministry of Public Security - plays a pivotal role in digitizing public services and connecting citizens with the state management system. As of February 2025, VNeID had accumulated over 55 million activated accounts and processed more than 2.1 billion document digitization transactions, demonstrating an immense deployment scale.

However, parallel to this impressive implementation scale are severe challenges regarding User Experience (UX): complex authentication workflows, non-intuitive interfaces, frequent technical errors, and accessibility barriers for vulnerable user groups such as the elderly and those with limited technological literacy. These limitations create a "digital divide," contradicting the Government's stated objective of "leaving no one behind." Driven by these practical realities, this study was conducted to comprehensively analyze the current UX issues of VNeID, propose enhanced design solutions, and evaluate their efficacy through controlled usability testing following a User-Centered Design (UCD) approach.

2. Scientific Content

2.1. Current State of the Research Problem

The research team collected data from three primary sources: (1) An online survey (Google Forms) involving 205 users across various age groups and technical proficiency levels; (2) Organic user feedback from social media platforms (Facebook, TikTok) comprising 162 user comments and complaints; (3) Application store reviews (App Store: 48,000 reviews, rating: 2.7/5; Google Play: 236,000 reviews, rating: 3.5/5).

The analysis reveals that VNeID's current UX issues are concentrated into three main clusters: (1) Usability: Multi-step verification processes (requiring passcode re-entry for every document viewing session), frequent login failures, and a lack of granular guidelines for administrative procedures. (2) UI/Content: Cluttered layouts, unintelligible information architecture, and visual inconsistency. (3) Accessibility/Navigation: Excessive operational steps, low discoverability of functions, and a complete lack of offline functionality.

Overall user satisfaction remains average (with a score of 3/5 representing the highest proportion at 33.7% in the survey), accompanied by a prolonged trend of negative sentiment stretching from 2023 to early 2026.

2.2. Research Methodology and Foundation

The study is constructed upon three core theoretical foundations: (1) The User-Centered Design (UCD) methodology in accordance with the ISO 9241-210 standard, utilizing the 5-stage Design Thinking framework: Empathize – Define – Ideate – Prototype – Test; (2) Internationally standardized user experience evaluation metrics: the System Usability Scale (SUS) and the User Experience Questionnaire (UEQ); (3) Design and prototyping software: Figma (for user interface design and high-fidelity prototyping of core user flows) combined with Framer (for complex interactive prototyping of the chatbot and offline modes), alongside Python/Google Colab and Microsoft Excel for data analysis and visualization.

2.3. Proposed Solutions and Research Findings

Based on the results of the situational analysis, the research team proposed and realized six solutions to improve the UX for VNeID:

(1) Optimizing the authentication mechanism: Replacing the mechanism that requires entering a passcode every time documents are viewed with a single-authentication mechanism for the entire group of secure functions within a certain session. After successful authentication, users can access all documents in the wallet without re-entering; the session automatically expires after a period of time to ensure security.

(2) Adding detailed guidelines in the form of a checklist: Placing the "View Guide" button right at the top interface of each administrative procedure (temporary residence registration, criminal record certificate issuance, citizen identity card renewal/reissuance), instead of being hidden deep within the Settings section. The checklist includes applicable subjects, documents to prepare, implementation steps, and fees.

(3) Draft-saving function and improving the Back button: The system automatically asks to save a draft when users exit midway; the Back button preserves the entered data, allowing users to continue completing the application later without losing information.

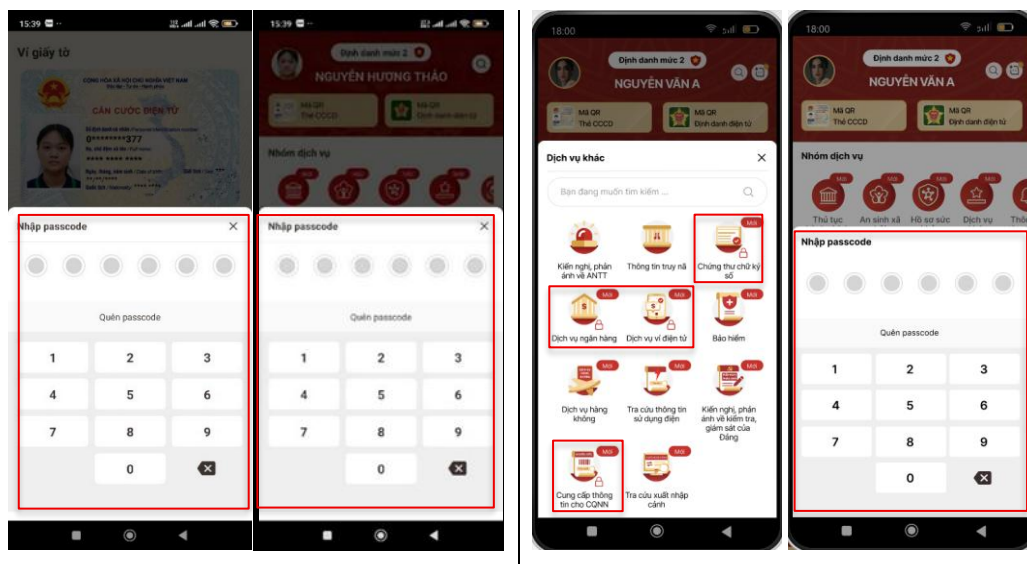
(4) Integrating a support chatbot: A direct chat frame within the application allows users to ask questions in natural language; the chatbot answers common situations and redirects to a support agent if it is outside the processing scope.

(5) Offline document viewing mode: Locally storing some basic information (ID card number, full name, date of birth, driver's license number, health insurance number) to help users view and present documents when there is no network connection.

(6) Tracking application status in a timeline format: Displaying the entire application processing progress chronologically with distinct color codes (yellow: processing, green: completed, red: rejected), accompanied by specific reasons if the application is rejected.

The prototype was built on Figma (the entire interface and main functional flows) and Framer (chatbot, offline mode). Qualitative testing results on 51 users showed that 82.4% proactively read the guidelines, 68.6% automatically gave praise, and 52.9% interacted with the "?" help icon. Quantitative testing results on 61 users gave an average SUS score of 88.25/100 – classified as "Excellent" according to the Bangor (2009) scale. UEQ results ranged from average to good, highest in the criteria of "Perspicuity" (4.5/7) and "Efficiency" (4.4/7). Particularly, the group of users with basic technology levels recorded the most impressive increase in SUS score (from 42.5 points with the current VNeID to 77.5 points with the improved prototype).

System Interface Visuals:



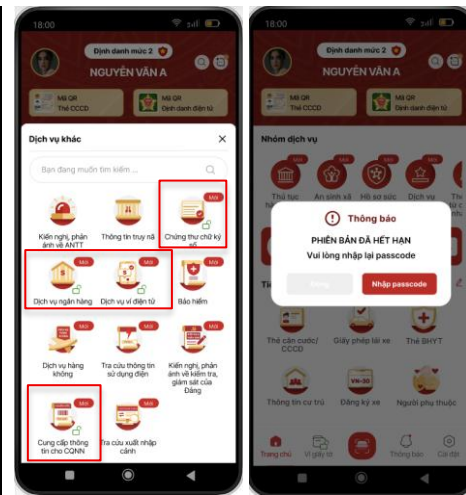


Figure 1: Authentication mechanism interface: current version (left) requires a passcode entry for each document view; enhanced version (right) features locked documents → single passcode entry → unlocks all documents with a session timeout limit.

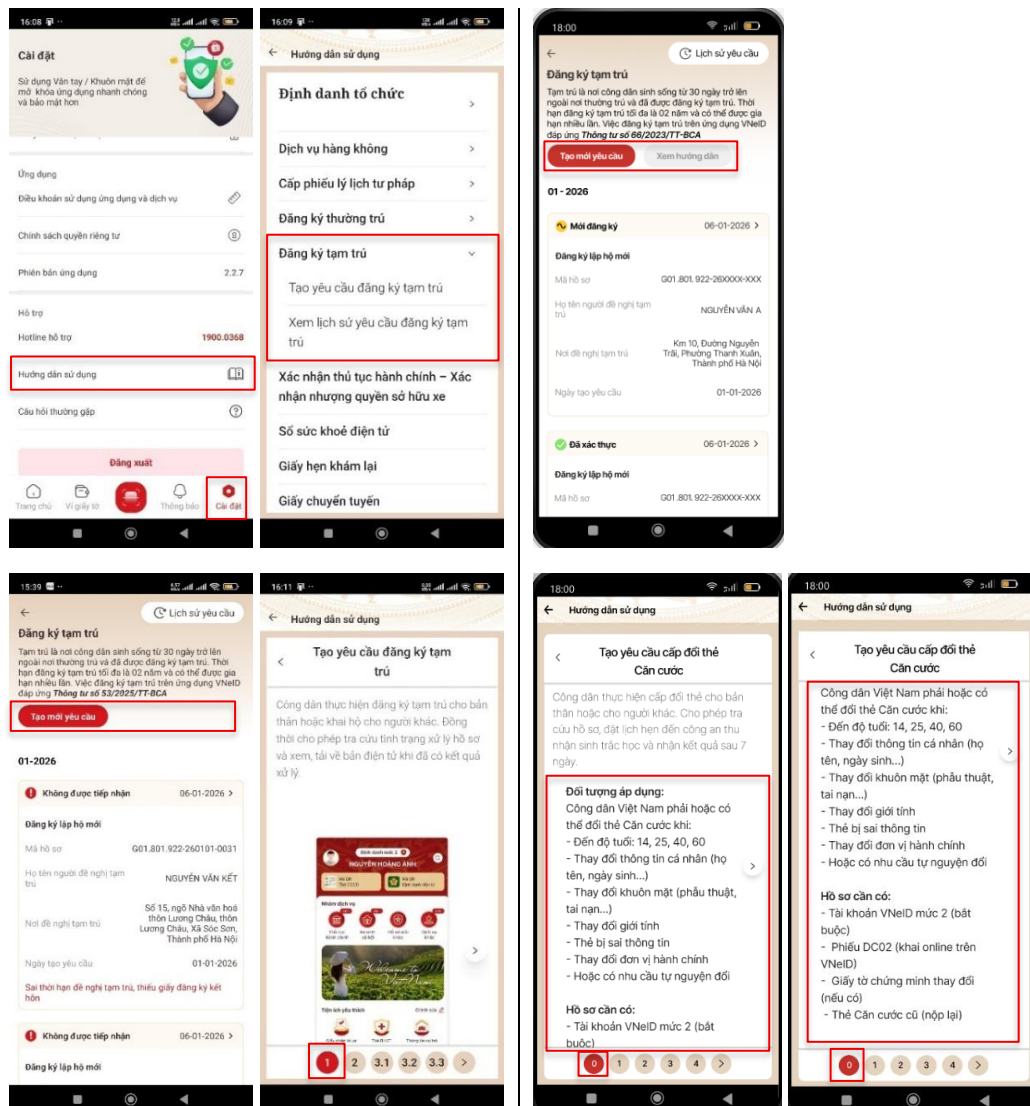


Figure 2: Procedural guidelines interface: current version (left) buries instructions deep inside Settings (red box); enhanced version (right) introduces a prominent "View Guide" button and a checklist at the header of each procedure.

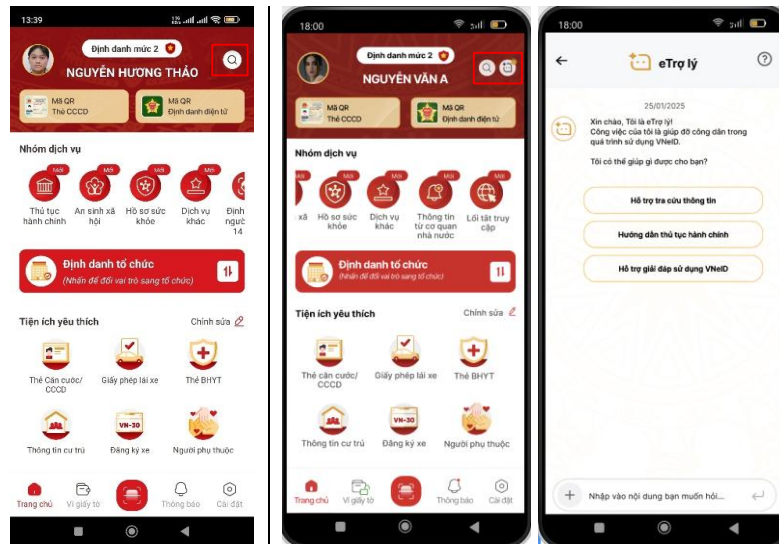


Figure 3: User assistance interface: current version (left) lacks direct support channels; enhanced version (right) integrates a support chatbot (red box) directly onto the primary interface.

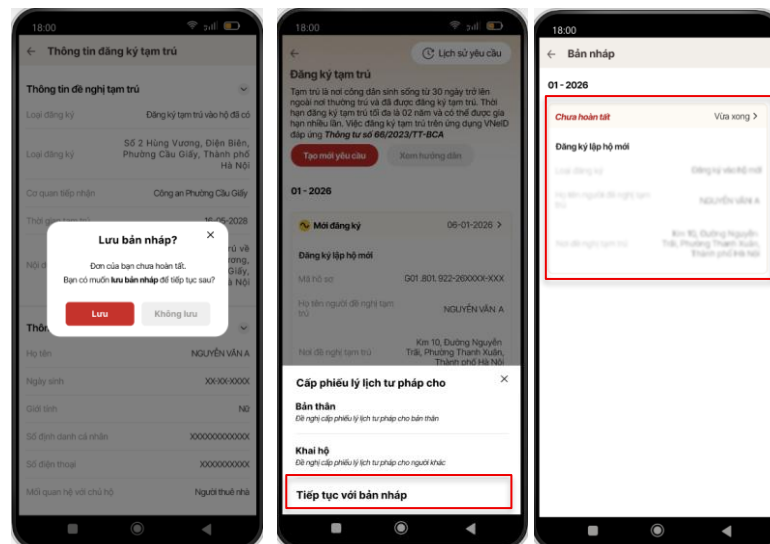
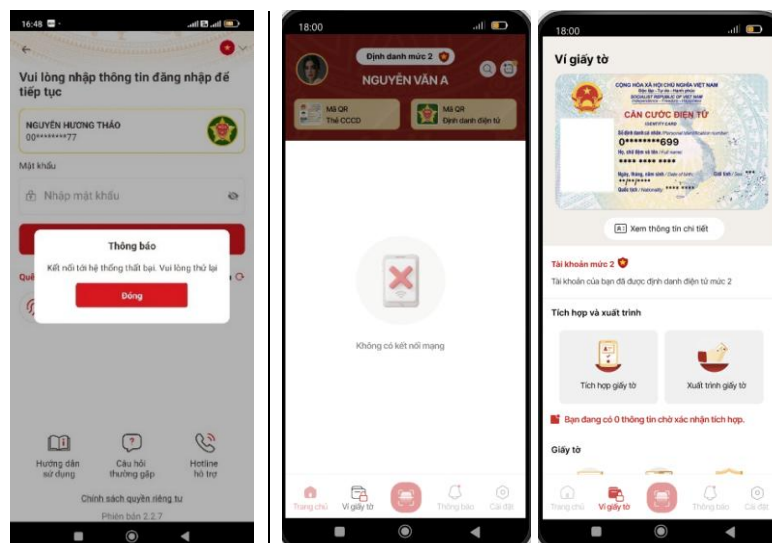


Figure 4: Draft-saving mechanism interface on the enhanced prototype.



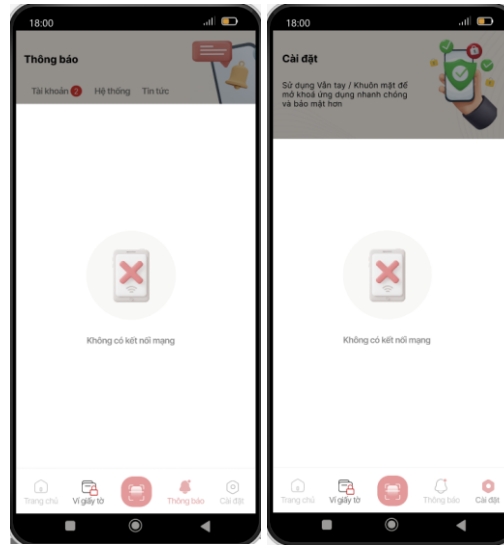


Figure 5: Offline mode interface: current version (left) provides no offline capabilities; enhanced version (right) allows the digital document wallet to display offline.

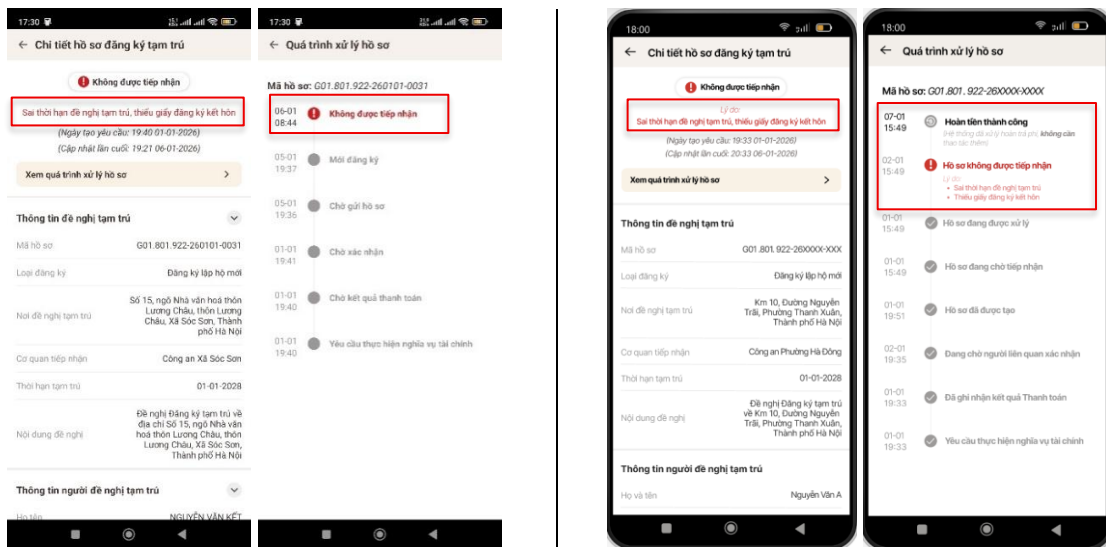


Figure 6: Application status tracking interface: current version (left); enhanced version (right) appends granular rejection justifications and introduces a "Refunded" status".

3. Conclusion and Recommendations

This study demonstrates that the application of User-Centered Design (UCD) principles is highly feasible and yields quantifiable efficacy in improving user experience for large-scale public administration applications like VNelD. Through systematic analysis, design iterations, and usability testing, the six proposed solutions effectively addressed the majority of core user pain points. Specifically, optimizations involving authentication workflow simplification, contextual guidance provisioning, and real-time user error support substantially amplified the system's overall usability and user satisfaction metrics.

To further iterate and scale the product in the future, the research team forwards the following targeted recommendations:

First: Expand the scope of usability testing on production-level environments with a larger sample size to ensure generalizability and data reliability, paying specific attention to the elderly and users in remote or rural areas who face steeper technological learning curves.

Second: Optimize the UI components toward age-inclusive design, specifically increasing the target sizing of icons and action buttons, while ensuring high color contrast and clear text legibility.

Third: Improve the discoverability of the "Drafts" feature by relocating it to a more prominent, intuitive position on the primary dashboard, ensuring users can readily identify and leverage it during drop-off events.

Fourth: Scale the chatbot architecture into an intelligent AI-driven virtual assistant capable of rendering automated predictive prompts during form-filling fields and offering real-time contextual assistance.

Fifth: Establish a specialized set of UX evaluation heuristics dedicated to public administrative applications, incorporating domain-specific factors such as process transparency, system trustworthiness, and legal reassurance during service consumption.

Finally: Extend cross-platform compatibility and implement multi-language support localized for foreign nationals, thereby enhancing the global accessibility and perception of Vietnam's digital public services.

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Governing Body

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